

Document Visualizer

RAG-Based Document Q&A with Automated Flowchart Generation

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Document Visualizer is an AI-powered tool that helps students and researchers understand research papers quickly and easily.

Users can upload any research paper or document and ask questions — the system retrieves the most relevant parts and provides clear, accurate answers with proper citations.

Simultaneously, it automatically extracts the methodology section and converts it into an **interactive, editable flowchart** using simple English instructions.

Built using a **hybrid retrieval** approach combined with **LLaMA-3**, and will be deployed as a free web app on Hugging Face Spaces.

Problem Statement

Reading research papers is slow and challenging:

- Papers are long (typically 8,000–12,000 words)
- Understanding methodology takes 3–5 hours
- Dense text without diagrams leads to misunderstandings
- Most AI tools provide only text answers (no visuals or quality checks)

Our Solution

Document Visualizer simplifies document comprehension by offering:

- Fast, accurate, citation-backed answers to natural language questions
- Automatic generation of editable flowcharts from the methodology section

1. Upload PDF

Drag & drop any research paper or document.

2. Document Processing

System parses, chunks, and indexes the document for retrieval.

3. Choose Tab

- **Q&A Tab** — Ask questions, get AI answers with source citations
- **Flowchart Tab** — Auto-generate flowcharts from detected process sections

4. Edit Flowchart

Modify the generated flowchart using plain English instructions — no diagram software needed.

5. View Sources

Expand any answer to see the exact referenced sections from the document.

Paper / Venue	Key Contribution	Limitations
Lewis et al. (NeurIPS 2020)	Introduced RAG — significantly reduces hallucination in QA	Fixed chunking; no visual output; limited scientific evaluation
Es et al. (arXiv 2023)	RAGAS framework for reference-free evaluation	Ignores scientific domain specifics
Cohan et al. (ACL 2020)	SPECTER embeddings excel on scientific retrieval	Static embeddings, not domain-fine-tuned
Guo et al. (SIGIR 2022)	Reciprocal Rank Fusion outperforms single retrievers	Heuristic fusion weights

Research Gap

No prior work integrates **hybrid retrieval (dense + sparse) + automated editable flowchart generation + RAGAS evaluation** specifically for scientific documents.

PDF Processing

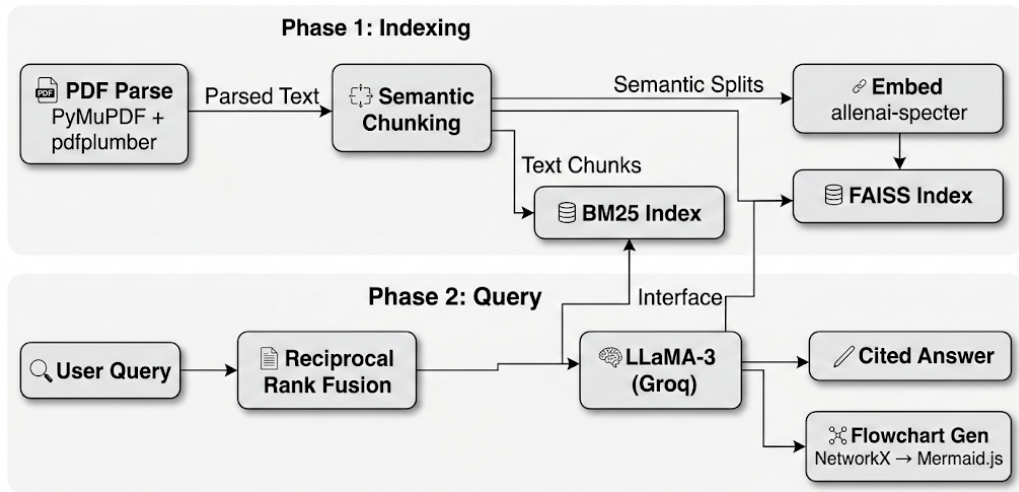
- PyMuPDF
- pdfplumber
- Tesseract OCR

Retrieval & Generation

- FAISS (semantic)
- BM25 (keyword)
- allenai-specter embeddings
- Groq + LLaMA-3 8B

App & Flowchart

- LangChain + Streamlit
- NetworkX + Mermaid.js
- RAGAS evaluation
- Hugging Face Spaces



Heavy Dependencies

The stack includes PyTorch (via `sentence-transformers`), FAISS, Tesseract OCR, and Groq — these are large and significantly slow down installation and cold-start times on Hugging Face Spaces.

Mermaid.js Rendering in Streamlit

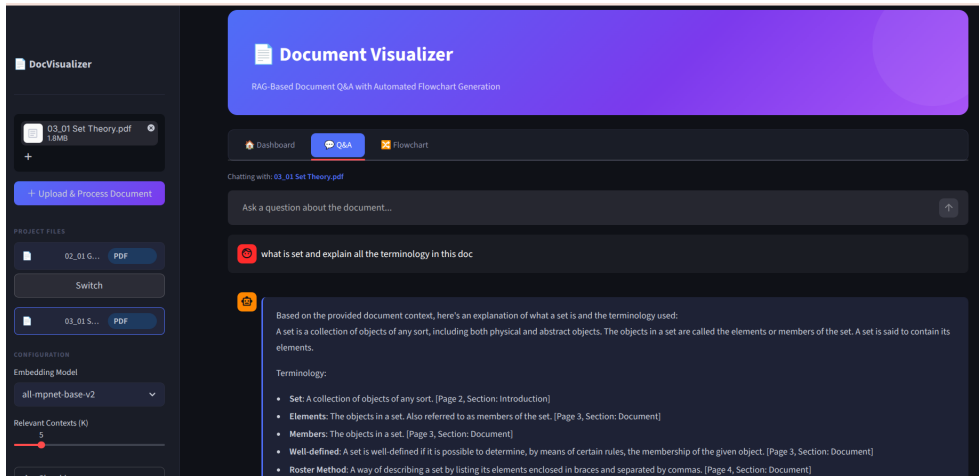
Streamlit has no native Mermaid support — rendering required injecting the library via `st.components.v1.html`. Fixed iframe height causes large flowcharts to be clipped or require scrolling, degrading the user experience.

Groq API Rate Limits

LLaMA-3 8B on the free tier enforces strict token-per-minute limits, causing request failures during rapid or back-to-back queries and requiring retry logic with exponential back-off.

OCR Reliability

Scanned PDFs with low DPI or skewed pages produce garbled text. Tesseract requires a minimum of 300 DPI for acceptable output; below this threshold, extracted text is unusable and Q&A quality degrades severely.



The screenshot displays the DocVisualizer application interface. On the left sidebar, there's a 'DocVisualizer' header, a file upload section showing '03_01 Set Theory.pdf' (1.8MB), and a '+ Upload & Process Document' button. Below this is a 'PROJECT FILES' section with a 'Switch' button and another file entry '03_01 S...'. The 'CONFIGURATION' section shows the 'Embedding Model' set to 'all-mpnet-base-v2' and 'Relevant Contexts (K)' set to 5. The main content area has a purple header 'Document Visualizer' with the subtitle 'RAG-Based Document Q&A with Automated Flowchart Generation'. It includes a navigation bar with 'Dashboard', 'Q&A' (active), and 'Flowchart' tabs. The 'Q&A' section shows 'Chatting with: 03_01 Set Theory.pdf' and a text input field 'Ask a question about the document...'. A question is entered: 'what is set and explain all the terminology in this doc'. The answer is displayed in a blue box: 'Based on the provided document context, here's an explanation of what a set is and the terminology used: A set is a collection of objects of any sort, including both physical and abstract objects. The objects in a set are called the elements or members of the set. A set is said to contain its elements. Terminology: • Set: A collection of objects of any sort. [Page 2, Section: Introduction] • Elements: The objects in a set. Also referred to as members of the set. [Page 3, Section: Document] • Members: The objects in a set. [Page 3, Section: Document] • Well-defined: A set is well-defined if it is possible to determine, by means of certain rules, the membership of the given object. [Page 3, Section: Document] • Roster Method: A way of describing a set by listing its elements enclosed in braces and separated by commas. [Page 4, Section: Document]'

Figure: RAG Pipeline: PDF upload → hybrid retrieval → cited answer

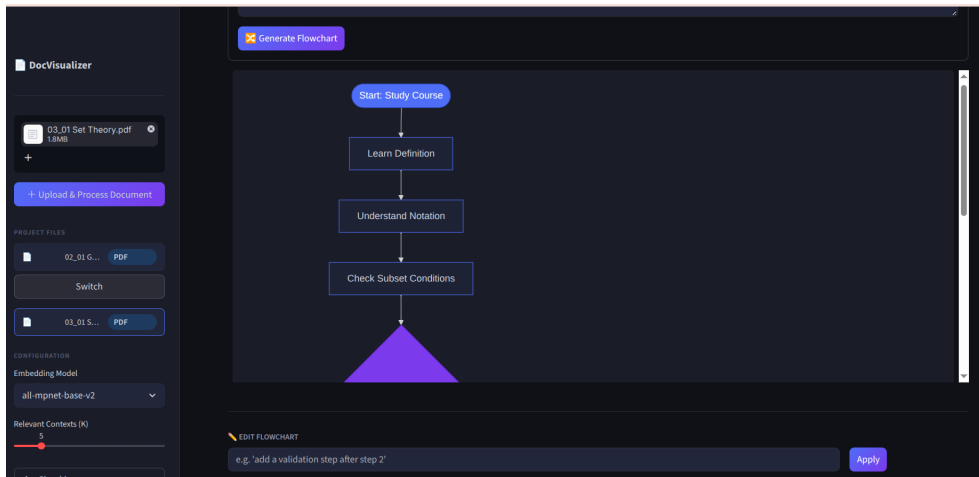


Figure: Flowchart Generation: Methodology detection → JSON → Interactive Mermaid diagram (editable)

Performance on a typical 12-page research paper

#	Metric	Value
1	PDF Parsing Speed	~3–5 s / page
2	Indexing Time	~1.2 s / 100 chunks
3	Query Response Time	~2–4 s end-to-end
4	Retrieval Hit Rate (Top-5)	~87%
5	Flowchart Generation	~3–6 s

Component	Status
PDF Parsing & Chunking	Complete
Hybrid Retrieval	Complete
Q&A with Citations	Complete
Flowchart Generation + Editing	Complete
RAGAS Evaluation	In Progress
Hugging Face Deployment	Pending

Education & Research

- Reduces comprehension time from hours to minutes
- Helps students, researchers in Tier-2/3 institutions
- Makes complex algorithms visually accessible

Inclusivity

100% free and open — no login or API key required. Works on research papers, legal, and technical documents.

Scientific Integrity

Every answer includes exact page + section citations — promotes evidence-based understanding and reduces misinformation.

Productivity at Scale

Extensible to medical, legal, and policy documents.

-  Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS 2020.
-  Es, S., et al. (2023). *RAGAS: Automated Evaluation of Retrieval Augmented Generation*. arXiv:2309.15217.
-  Cohan, A., et al. (2020). *SPECTER: Document-level Representation Learning using Citation-informed Transformers*. ACL 2020.
-  Guo, R., et al. (2022). *Efficient Hybrid Retrieval with Reciprocal Rank Fusion*. SIGIR 2022.

Thank You

Questions & Feedback Welcome